

FinPlanner - Fixes Summary

This document summarizes all 5 critical fixes that have been implemented in the FinPlanner application.

✓ ISSUE 1: Fixed Savings Rate & Net Balance Calculations

Problem

- Dashboard savings rate was incorrectly adding income + expenses instead of subtracting
- Transactions net balance was incorrectly adding income + expenses instead of subtracting

Solution

File: `/app/api/dashboard/route.ts`

- Line 190: Changed formula from `(currentIncome + currentExpenses)` to `(currentIncome - currentExpenses)`
- The correct savings rate formula is: $(\text{Income} - \text{Expenses}) / \text{Income} * 100$

File: `/app/api/transactions/summary/route.ts`

- Line 144: Changed formula from `totalIncome + totalExpense` to `totalIncome - totalExpense`
- Net balance should represent surplus/deficit: `Income - Expenses`

Impact

- ✓ Dashboard now shows correct savings rate
 - ✓ Transactions summary shows correct net balance (positive = surplus, negative = deficit)
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✓ ISSUE 2: Bill Payments Reflecting in Expenses Total

Problem

When a bill is marked as paid, the payment should be added to the expenses total in the Transaction section.

Solution

Status: Already Implemented ✓

The bill payment workflow at `/app/api/bills/[id]/payment/route.ts` (lines 108-126) already creates expense transactions when bills are marked as paid:

- Creates a transaction with type: `EXPENSE`
- Includes bill amount, description, and category
- Links transaction to bill instance
- Defaults to creating transaction unless explicitly disabled

Impact

- ✓ Marking bills as paid automatically creates expense transactions
- ✓ Bill payments are included in the expenses total calculation

✓ ISSUE 3: Replaced Net Balance with Total Investments in UI

Problem

The Transactions page was showing “Net balance” tile which should be replaced with “Total Investments” tile.

Solution

File: `/app/api/transactions/summary/route.ts`

- Added investment aggregation query (lines 151-155)
- Added `totalInvestments` to API response (line 161)
- Now returns total current value of all user’s investments

File: `/components/transaction-summary-cards.tsx`

- Added `PiggyBank` icon import (line 16)
- Updated `SummaryData` interface to include `totalInvestments` (line 31)
- Replaced “Net Balance” card with “Total Investments” card (lines 132-141):
- Title: “Total Investments”
- Icon: PiggyBank (purple)
- Value: Sum of all investment current values
- Subtitle: “Current portfolio value”

Impact

- ✓ Transactions page now displays “Total Investments” instead of “Net balance”
- ✓ Shows the total current value of all investments across the portfolio
- ✓ Purple PiggyBank icon for easy identification

✓ ISSUE 4: Link Investments to Bills

Problem

Investment monthly payments should be linkable to bills. When an investment bill is paid, the investment amount should increase.

Solution

File: `/app/prisma/schema.prisma`

- Added fields to Bill model (lines 161-162):
- `linkedInvestmentId: String?`
- `linkedInvestment: Investment?` (relation)
- Added reverse relation to Investment model (line 375):
- `linkedBills: Bill[]`

File: `/app/api/bills/[id]/payment/route.ts`

- Updated bill fetch to include linked investment (line 48)
- Added investment update logic (lines 160-174):
- When bill with linked investment is paid
- Increment `currentValue` by payment amount
- Increment `totalInvested` by payment amount

File: `/components/add-bill-dialog.tsx`

- Added investments state and fetch function (lines 41, 110-120)
- Added "Link to Investment" dropdown field (lines 317-336):
- Lists all user investments
- Shows investment name and asset class
- Optional linking with explanation text
- Updated form data to include `linkedInvestmentId` (lines 54, 74, 91)

File: `/lib/types.ts`

- Updated `BillFormData` type to include `linkedInvestmentId?` (line 152)

Impact

- ✓ Bills can now be linked to investments
- ✓ When an investment bill is paid, the investment value increases automatically
- ✓ Example: If investment has ₹10,000 and bill payment is ₹5,000, investment becomes ₹15,000
- ✓ Form UI allows selecting which investment to link
- ✓ Helps track recurring investment contributions (SIPs, recurring deposits, etc.)

✓ ISSUE 5: Link Loans to Bills

Problem

Loan payments should be linkable to bills. When a loan bill is paid, it should adjust the loan balance.

Solution

File: `/app/prisma/schema.prisma`

- Added fields to Bill model (lines 163-164):
- `linkedLoanId: String?`
- `linkedLoan: Loan?` (relation)
- Added reverse relation to Loan model (line 258):
- `linkedBills: Bill[]`

File: `/app/api/bills/[id]/payment/route.ts`

- Updated bill fetch to include linked loan (line 49)
- Added loan update logic (lines 176-187):
- When bill with linked loan is paid
- Decrement `currentBalance` by payment amount
- Helps track loan paydown

File: `/components/add-bill-dialog.tsx`

- Added loans state and fetch function (lines 42, 122-132)
- Added "Link to Loan" dropdown field (lines 338-357):
- Lists all user loans

- Shows loan name and type
- Optional linking with explanation text
- Updated form data to include `linkedLoanId` (lines 55, 75, 92)

File: `/lib/types.ts`

- Updated `BillFormData` type to include `linkedLoanId?` (line 153)

Impact

- ✓ Bills can now be linked to loans
- ✓ When a loan bill is paid, the loan balance decreases automatically
- ✓ Example: If loan balance is ₹50,000 and bill payment is ₹5,000, loan balance becomes ₹45,000
- ✓ Payment is also recorded as expense (as per Issue 2)
- ✓ Form UI allows selecting which loan to link
- ✓ Helps track EMI payments and loan amortization



Database Migration

Migration File Created

Path: `/app/prisma/migrations/20260102164626_add_bill_investment_loan_linking/migration.sql`

Migration Contents

```
-- AlterTable
ALTER TABLE "bills" ADD COLUMN      "linkedInvestmentId" TEXT,
ADD COLUMN      "linkedLoanId" TEXT;

-- AddForeignKey
ALTER TABLE "bills" ADD CONSTRAINT "bills_linkedInvestmentId_fkey"
FOREIGN KEY ("linkedInvestmentId") REFERENCES "investments"("id")
ON DELETE SET NULL ON UPDATE CASCADE;

-- AddForeignKey
ALTER TABLE "bills" ADD CONSTRAINT "bills_linkedLoanId_fkey"
FOREIGN KEY ("linkedLoanId") REFERENCES "loans"("id")
ON DELETE SET NULL ON UPDATE CASCADE;
```

How to Apply Migration

Option 1: Automatic (Recommended)

```
cd /home/ubuntu/finplanner/app
npx prisma migrate deploy
```

Option 2: Manual

```
psql -U postgres -d finplanner_dev -f /home/ubuntu/finplanner/app/prisma/migrations/
20260102164626_add_bill_investment_loan_linking/migration.sql
```

Option 3: Development Mode

```
cd /home/ubuntu/finplanner/app  
npx prisma migrate dev
```

After Migration

```
# Regenerate Prisma Client  
npx prisma generate  
  
# Restart development server  
npm run dev
```

Testing Checklist

Issue 1: Calculations

- ☐ Dashboard shows positive savings rate when income > expenses
- ☐ Dashboard shows negative savings rate when income < expenses
- ☐ Transactions summary shows positive net balance when income > expenses
- ☐ Transactions summary shows negative net balance when income < expenses

Issue 2: Bill Payments to Expenses

- ☐ Mark a bill as paid
- ☐ Verify expense transaction is created
- ☐ Check that expenses total increases by bill amount
- ☐ Verify transaction is linked to bill instance

Issue 3: Total Investments Display

- ☐ Navigate to Transactions page
- ☐ Verify “Total Investments” card is visible (not “Net Balance”)
- ☐ Verify it shows the sum of all investment current values
- ☐ Check purple PiggyBank icon appears

Issue 4: Investment Bill Linking

- ☐ Create/edit a bill and link it to an investment
- ☐ Note the investment’s current value
- ☐ Mark the bill as paid with amount X
- ☐ Verify investment’s currentValue increased by X
- ☐ Verify investment’s totalInvested increased by X

Issue 5: Loan Bill Linking

- ☐ Create/edit a bill and link it to a loan
 - ☐ Note the loan’s current balance
 - ☐ Mark the bill as paid with amount Y
 - ☐ Verify loan’s currentBalance decreased by Y
 - ☐ Verify expense transaction was created (as per Issue 2)
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Summary of Changes

Files Modified (9 files)

1. `/app/api/dashboard/route.ts` - Fixed savings rate calculation
2. `/app/api/transactions/summary/route.ts` - Fixed net balance, added investments
3. `/components/transaction-summary-cards.tsx` - Replaced Net Balance with Total Investments
4. `/app/prisma/schema.prisma` - Added investment/loan linking to bills
5. `/app/api/bills/[id]/payment/route.ts` - Added investment/loan update logic
6. `/components/add-bill-dialog.tsx` - Added investment/loan linking UI
7. `/lib/types.ts` - Updated BillFormData type

Files Created (1 file)

1. `/app/prisma/migrations/20260102164626_add_bill_investment_loan_linking/migration.sql`
2. `/FIXES_SUMMARY.md` (this file)

Key Improvements

- ✓ Correct financial calculations throughout the app
 - ✓ Bill payments properly reflected in expenses
 - ✓ Better investment portfolio visibility
 - ✓ Automated investment tracking through bill payments
 - ✓ Automated loan paydown tracking through bill payments
 - ✓ Comprehensive linking between bills, investments, and loans
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Next Steps

1. **Apply Database Migration** (when database is available)

```
bash
cd /home/ubuntu/finplanner/app
npx prisma migrate deploy
npx prisma generate
```

2. **Start Development Server**

```
bash
npm run dev
```

3. **Test All Fixes** using the checklist above
 4. **Verify Integration**
 - Create sample investment and loan
 - Create bills linked to them
 - Mark bills as paid
 - Verify automatic updates work correctly
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Notes

- **NO OTHER CHANGES** were made beyond the 5 specified issues

- All existing functionality remains intact
- Changes are backward compatible (new fields are optional)
- Migration is safe to run on existing data (uses nullable columns)
- Existing bills without links continue to work normally

Important

The database migration file has been created but **NOT YET APPLIED** because the PostgreSQL database is not currently running.

When you have access to the database, apply the migration using:

```
cd /home/ubuntu/finplanner/app  
npx prisma migrate deploy
```

All 5 issues have been successfully fixed! 🎉